CHEMISTRY STUDY GUIDE

Fundamental Concepts and Principles

1. Atomic Structure

Chemistry is the scientific study of matter, its properties, how and why substances combine or separate to form other substances, and how substances interact with energy.

Key Historical Developments:
• Dalton's Atomic Theory (1803): All matter composed of atoms
• Mendeleev's Periodic Table (1869): Organized elements by properties
• Bohr's Atomic Model (1913): Electrons in discrete energy levels

The atom consists of:
- Protons (+ charge): Located in nucleus, mass ≈ 1 amu
- Neutrons (neutral): Located in nucleus, mass ≈ 1 amu
- Electrons (- charge): Orbit nucleus, mass ≈ 1/1836 amu

Atomic Number and Mass Number:

Mass Number (A) = Protons + Neutrons

Atomic Number (Z) = Number of Protons

Common Elements:

Element	Symbol	Atomic #	Atomic Mass
Hydrogen	H	1	1.008
Carbon	C	6	12.011
Oxygen	O	8	15.999
Sodium	Na	11	22.990
Chlorine	Cl	17	35.453

CHEMICAL BONDING & REACTIONS

2. Types of Chemical Bonds

Chemical bonds are forces that hold atoms together in molecules and compounds. Three primary types exist:

1. IONIC BONDS:
- Formed by complete transfer of electrons
- Between metals and non-metals
- Example: NaCl (Sodium Chloride)
- High melting/boiling points

2. COVALENT BONDS:
- Formed by sharing electron pairs
- Between non-metals
- Example: H■O (Water), CH■ (Methane)
- Can be polar or non-polar

3. METALLIC BONDS:
- Sea of delocalized electrons
- Between metal atoms
- Example: Cu (Copper), Fe (Iron)
- Good conductors of heat/electricity

3. Chemical Equations

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Balanced Equation Example:

  2H■ + O■ → 2H■O
  (Hydrogen + Oxygen → Water)

Types of Reactions:

1. Combination: A + B → AB
   2Mg + O■ → 2MgO

2. Decomposition: AB → A + B
   2H■O■ → 2H■O + O■

3. Single Replacement: A + BC → AC + B
   Zn + 2HCl → ZnCl■ + H■

4. Double Replacement: AB + CD → AD + C
   AgNO■ + NaCl → AgCl + NaNO■
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Alkali Metals	Halogens

Periodic Table Trends:

- Atomic Radius: Increases ↓, decreases →
- Ionization Energy: Increases →, decreases ↓
- Electronegativity: Increases →, decreases ↓
- Metallic Character: Increases ↓, decreases →
- Electron Affinity: Generally increases →

4. Organic Chemistry Fundamentals

Organic chemistry studies carbon-containing compounds.
Carbon's unique properties:
• Forms 4 covalent bonds
• Can form chains, rings, and complex structures
• Tetravalent nature allows for isomers

Hydrocarbon Classes:
1. Alkanes: Single bonds (C-C), saturated
   Example: Methane (CH■), Ethane (C■H■)

2. Alkenes: Double bonds (C=C), unsaturated
   Example: Ethene (C■H■)

3. Alkynes: Triple bonds (C≡C), unsaturated
   Example: Ethyne (C■H■)

4. Aromatic: Benzene rings (C■H■)
   Delocalized π electrons

Common Functional Groups:

Group	Formula	Example	Properties
Alcohol	-OH	C■H■OH	Polar, hydrogen bond
Aldehyde	-CHO	CH■O	Reactive, carbonyl
Ketone	C=O	CH■COCH■	Solvent properties
Carboxylic Acid	-COOH	CH■COOH	Acidic, hydrogen bon
Amine	-NH■	CH■NH■	Basic, forms salts

5. Biochemistry Essentials

MACROMOLECULES OF LIFE:

• Carbohydrates: Energy source (C■H■■O■ - Glucose)

• Lipids: Energy storage, cell membranes

• Proteins: Structure, enzymes (amino acid polymers)

• Nucleic Acids: Genetic information (DNA, RNA)

Key Biochemical Reactions:

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Photosynthesis: 6CO■ + 6H■O + light → C■H■■O■ + 6O■
Cellular Respiration: C■H■■O■ + 6O■ → 6CO■ + 6H■O + ATP
Hydrolysis: Breaking bonds with water
Dehydration Synthesis: Forming bonds, releasing water
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6. Solutions and Concentrations

Solution = Solute + Solvent

Concentration Units:
• Molarity (M) = moles solute / liters solution
• Molality (m) = moles solute / kg solvent
• Mass % = (mass solute / mass solution) × 100%
• ppm = (mass solute / mass solution) × 10■

Solubility Rules:
1. All Na■, K■, NH■■ salts are soluble
2. All NO■■, ClO■■, ClO■■ salts are soluble
3. Most Cl■, Br■, I■ salts soluble (except Ag■, Pb²■, Hg■²■)
4. Most SO■²■ salts soluble (except Ba²■, Pb²■, Ca²■)
5. Most OH■ salts insoluble (except Group 1, NH■■, Ba²■)
6. Most S²■, CO■²■, PO■³■ salts insoluble

7. Chemical Thermodynamics

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Laws of Thermodynamics:

1st Law: ΔU = Q - W
   (Energy conservation)

2nd Law: ΔS_universe ≥ 0
   (Spontaneous processes increase disorder)

3rd Law: S → 0 as T → 0 K
   (Perfect crystal at 0 K has zero entropy)

Gibbs Free Energy: ΔG = ΔH - TΔS
• ΔG < 0: Spontaneous
• ΔG > 0: Non-spontaneous
• ΔG = 0: Equilibrium
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8. Chemical Kinetics

Rate Laws:

For aA + bB → products

`Rate = k[A]^m[B]^n`

Arrhenius Equation:

`k = A·e^(-Ea/RT)`

where:

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k = rate constant
A = frequency factor
Ea = activation energy
R = gas constant
T = temperature (K)
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9. Gas Laws

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Ideal Gas Law: PV = nRT
R = 0.0821 L·atm/mol·K

Combined Gas Law: P■V■/T■ = P■V■/T■

Dalton's Law: P_total = ΣP_i

Graham's Law: Rate■/Rate■ = √(M■/M■)
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Van der Waals Equation:

`[P + a(n/V)²](V - nb) = nRT`

Key Takeaways:

1. Chemistry connects microscopic particles to macroscopic properties
2. Periodic trends help predict element behavior
3. Bond type determines compound properties
4. Organic compounds form basis of life
5. Energy changes drive chemical reactions
6. Equilibrium balances forward and reverse reactions